

MERN Stack Deployment Guide for Hostinger VPS (Ubuntu 22.04)

Prerequisites

- Hostinger VPS with Ubuntu 22.04
- SSH access to your VPS
- Domain name (optional but recommended)
- Your MERN application source code

Step 1: Initial Server Setup

Connect to your VPS

```
bash
```

```
ssh root@your-server-ip
```

Update the system

```
bash
```

```
sudo apt update && sudo apt upgrade -y
```

Common Error: Permission denied **Solution:** Ensure you're using the correct SSH key or password provided by Hostinger

Create a new user (security best practice)

```
bash
```

```
adduser yourusername
```

```
usermod -aG sudo yourusername
```

Switch to the new user

```
bash
```

```
su - yourusername
```

Step 2: Install Node.js and npm

Install Node.js using NodeSource repository

```
bash
```

```
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -  
sudo apt-get install -y nodejs
```

Verify installation

```
bash
```

```
node --version  
npm --version
```

Common Error: `curl: command not found` **Solution:**

```
bash
```

```
sudo apt install curl -y
```

Common Error: Node.js version issues **Solution:** Use Node Version Manager (nvm)

```
bash
```

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.0/install.sh | bash  
source ~/.bashrc  
nvm install 18  
nvm use 18
```

Step 3: Install MongoDB

Import MongoDB public key

```
bash
```

```
wget -qO - https://www.mongodb.org/static/pgp/server-6.0.asc | sudo apt-key add -
```

Add MongoDB repository

```
bash
```

```
echo "deb [ arch=amd64,arm64 ] https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/6.0 "
```



Install MongoDB

```
bash
```

```
sudo apt update  
sudo apt install -y mongodb-org
```

Start and enable MongoDB

```
bash
```

```
sudo systemctl start mongod  
sudo systemctl enable mongod
```

Verify MongoDB installation

```
bash
```

```
sudo systemctl status mongod
```

Common Error: MongoDB fails to start **Solution:** Check logs and fix permissions

```
bash
```

```
sudo journalctl -u mongod  
sudo chown -R mongodb:mongodb /var/lib/mongodb  
sudo chown mongodb:mongodb /tmp/mongodb-27017.sock  
sudo service mongod restart
```

Common Error: `wget: command not found` **Solution:**

```
bash
```

```
sudo apt install wget -y
```

Step 4: Install Git and Clone Your Repository

Install Git

```
bash
```

```
sudo apt install git -y
```

Clone your repository

```
bash
```

```
cd /home/yourusername  
git clone https://github.com/yourusername/your-mern-app.git  
cd your-mern-app
```

Common Error: Permission denied (publickey) **Solution:** Set up SSH keys for GitHub or use HTTPS

```
bash
```

```
git clone https://github.com/yourusername/your-mern-app.git
```

Step 5: Set Up Your Application

Install backend dependencies

```
bash
```

```
cd backend # or wherever your server code is  
npm install
```

Install frontend dependencies

```
bash
```

```
cd ../frontend # or wherever your client code is  
npm install
```

Build the React application

```
bash
```

```
npm run build
```

Common Error: `npm ERR! peer dep missing` **Solution:**

```
bash
```

```
npm install --legacy-peer-deps
```

Common Error: Build fails due to memory issues **Solution:** Increase swap space

```
bash
```

```
sudo fallocate -l 2G /swapfile  
sudo chmod 600 /swapfile  
sudo mkswap /swapfile  
sudo swapon /swapfile  
echo '/swapfile none swap sw 0 0' | sudo tee -a /etc/fstab
```

Step 6: Configure Environment Variables

Create environment file for backend

```
bash
```

```
cd backend  
nano .env
```

Add your environment variables

```
NODE_ENV=production  
PORT=5000  
MONGODB_URI=mongodb://localhost:27017/your-database-name  
JWT_SECRET=your-jwt-secret
```

Common Error: Environment variables not loading **Solution:** Install dotenv and ensure proper path

```
bash
```

```
npm install dotenv
```

Step 7: Install and Configure PM2 (Process Manager)

Install PM2 globally

```
bash
```

```
sudo npm install -g pm2
```

Start your backend with PM2

```
bash
```

```
cd backend  
pm2 start server.js --name "mern-backend"
```

Configure PM2 to start on boot

```
bash
```

```
pm2 startup  
pm2 save
```

Common Error: PM2 command not found **Solution:** Add npm global bin to PATH

```
bash
```

```
echo 'export PATH=$PATH:/usr/lib/node_modules/npm/bin' >> ~/.bashrc  
source ~/.bashrc
```

Step 8: Install and Configure Nginx

Install Nginx

```
bash
```

```
sudo apt install nginx -y
```

Start and enable Nginx

```
bash
```

```
sudo systemctl start nginx  
sudo systemctl enable nginx
```

Configure Nginx for your MERN app

```
bash
```

```
sudo nano /etc/nginx/sites-available/your-domain.com
```

Add this configuration:

nginx

```
server {  
    listen 80;  
    server_name your-domain.com www.your-domain.com;  
  
    # Serve React build files  
    location / {  
        root /home/yourusername/your-mern-app/frontend/build;  
        index index.html index.htm;  
        try_files $uri $uri/ /index.html;  
    }  
  
    # Proxy API requests to backend  
    location /api {  
        proxy_pass http://localhost:5000;  
        proxy_http_version 1.1;  
        proxy_set_header Upgrade $http_upgrade;  
        proxy_set_header Connection 'upgrade';  
        proxy_set_header Host $host;  
        proxy_set_header X-Real-IP $remote_addr;  
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
        proxy_set_header X-Forwarded-Proto $scheme;  
        proxy_cache_bypass $http_upgrade;  
    }  
}
```

Enable the site

bash

```
sudo ln -s /etc/nginx/sites-available/your-domain.com /etc/nginx/sites-enabled/  
sudo nginx -t  
sudo systemctl reload nginx
```

Common Error: Nginx configuration test fails **Solution:** Check syntax and fix issues

bash

```
sudo nginx -t  
# Fix any syntax errors shown
```

Common Error: 502 Bad Gateway **Solution:** Ensure backend is running and check PM2 status

```
bash
```

```
pm2 status
```

```
pm2 logs
```

Step 9: Configure Firewall

Set up UFW firewall

```
bash
```

```
sudo ufw allow OpenSSH
```

```
sudo ufw allow 'Nginx Full'
```

```
sudo ufw enable
```

Check firewall status

```
bash
```

```
sudo ufw status
```

Step 10: Set Up SSL Certificate (Optional but Recommended)

Install Certbot

```
bash
```

```
sudo apt install certbot python3-certbot-nginx -y
```

Obtain SSL certificate

```
bash
```

```
sudo certbot --nginx -d your-domain.com -d www.your-domain.com
```

Common Error: Domain verification fails **Solution:** Ensure your domain points to your server IP and DNS has propagated

Step 11: Database Setup

Access MongoDB shell

```
bash
```

```
mongosh
```


Create database and user

```
javascript

use your-database-name
db.createUser({
  user: "your-username",
  pwd: "your-password",
  roles: [{ role: "readWrite", db: "your-database-name" }]
})
```

Update your .env file with authentication

```
MONGODB_URI=mongodb://your-username:your-password@localhost:27017/your-database-name
```

Step 12: Final Verification

Check all services are running

```
bash

sudo systemctl status nginx
sudo systemctl status mongod
pm2 status
```

Test your application

- Visit your domain in a browser
- Test API endpoints
- Check database connections

Common Troubleshooting Commands

Check logs

```
bash
```

```
# Nginx Logs
```

```
sudo tail -f /var/log/nginx/error.log
```

```
sudo tail -f /var/log/nginx/access.log
```

```
# PM2 Logs
```

```
pm2 logs
```

```
# MongoDB Logs
```

```
sudo journalctl -u mongod
```

```
# System Logs
```

```
sudo journalctl -xe
```

Restart services

```
bash
```

```
sudo systemctl restart nginx
```

```
sudo systemctl restart mongod
```

```
pm2 restart all
```

Monitor resources

```
bash
```

```
htop
```

```
df -h
```

```
free -h
```

Additional Security Measures

Change default SSH port

```
bash
```

```
sudo nano /etc/ssh/sshd_config
```

```
# Change Port 22 to Port 2222
```

```
sudo systemctl restart ssh
```

Configure MongoDB security

```
bash
```

```
sudo nano /etc/mongod.conf
# Add:
# security:
#   authorization: enabled
sudo systemctl restart mongod
```

Regular updates

```
bash
```

```
sudo apt update && sudo apt upgrade -y
npm update
```

Backup Strategy

Database backup

```
bash
```

```
mongodump --db your-database-name --out /home/yourusername/backups/
```

Application backup

```
bash
```

```
tar -czf /home/yourusername/backups/app-backup-$(date +%Y%m%d).tar.gz /home/yourusername/
```



Performance Optimization

Enable Gzip compression in Nginx

```
nginx
```

```
# Add to your nginx config
gzip on;
gzip_vary on;
gzip_min_length 1024;
gzip_types text/plain text/css text/xml text/javascript application/javascript application/javascript
```



Optimize MongoDB

```
bash
```

```
# Edit MongoDB config
```

```
sudo nano /etc/mongod.conf
```

```
# Add appropriate settings for your server resources
```

This guide covers the complete deployment process. Remember to replace placeholder values with your actual configuration details and always test each step before moving to the next one.